

Bubble-UI: A Fatigue-Reducing and User-Friendly Spatial UI System Using a Soap Bubble Metaphor

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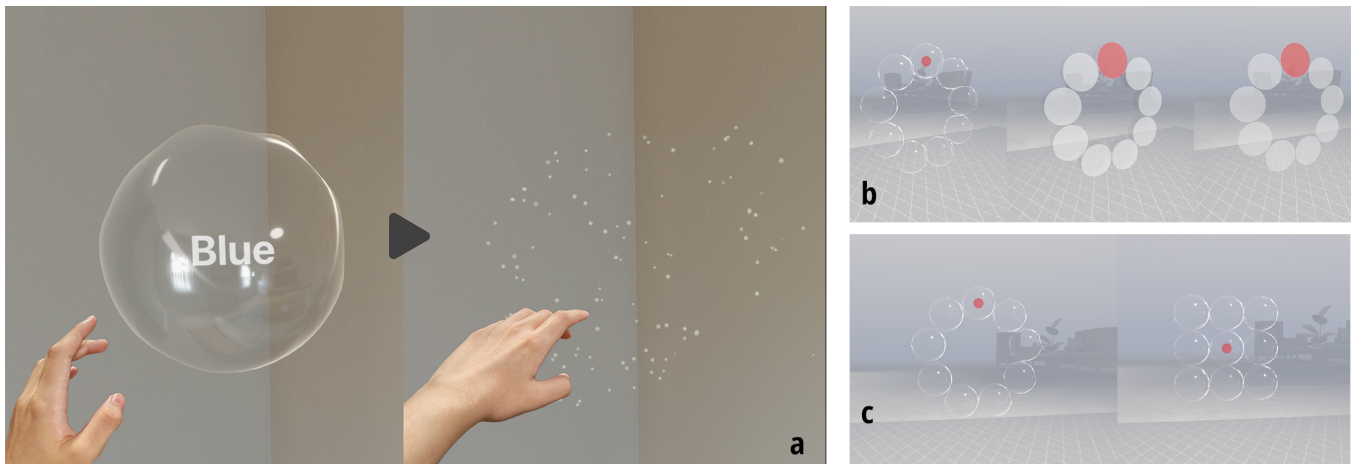


Figure 1: (a) Bubble-Button: A spatial UI system using soap bubbles as a metaphor operating on visionOS. It bursts on finger touch to register selection. (b) The Poke and Touch buttons compared in the experiment. (c) The layout of the two types of tasks used in the experiment.

Abstract

In spatial UI systems used in VR and AR, hardware interfaces are widely employed as metaphors. Utilizing metaphors offers advantages such as facilitating the construction of appropriate mental models for behavior and enabling accurate operations. However, in existing spatial UI metaphors, operability is often sacrificed. In this research, we propose Bubble-UI, which utilizes soap bubbles as a UI metaphor, aiming to enhance comprehensibility and improve performance by leveraging interaction knowledge. As applications, we propose three types: "Bubble-Button" for button operations, "Bubble-Holder" for object manipulation, and "Bubble-Loading" for loading representations. Among these, experimental results for Bubble-Button indicate that it is faster in input speed, less tiring,

and more attractive compared to other methods, while exhibiting a higher error rate.

CCS Concepts

• **Human-centered computing** → **Gestural input**; *Interaction design theory, concepts and paradigms*; Empirical studies in interaction design.

Keywords

Virtual reality, Mid-air interaction, Target selection, Crossing paradigm, Object manipulation, Metaphor

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1 Introduction

Metaphors are a rhetorical device that promotes understanding by likening one thing to another, playing a crucial role in user interface (UI) design. By incorporating metaphors into UIs, users can

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more easily understand new systems and operation methods by associating them with their existing knowledge and experiences. For example, Microsoft Windows and Apple macOS facilitate understanding of the file system by using the desktop as a metaphor.

The appropriate use of metaphors in UIs facilitates the construction of mental models for interface behavior and improves learnability [18]. Furthermore, appropriate metaphors enable more accurate operations [12]. In the field of manipulating objects using 3D UIs, how to set operation metaphors is discussed as an important factor [11, 16, 21].

In XR environment UIs, existing mechanical interfaces are often used as metaphors [3, 4, 6]. Using existing hardware metaphors allows users to leverage their interaction knowledge, making it easier to construct mental models. If the hardware metaphor is familiar, users can directly apply their interaction knowledge.

However, using existing hardware metaphors poses the problem of operations not being optimized for XR spaces.

The characteristics of XR spaces include the unnecessary of physical mechanisms for judgment, the ability to negate gravity, and the selectable presence of haptic feedback. A problem when using existing hardware as a metaphor is that the constraints on operations created by physical mechanisms are also carried over. For example, those that reproduce mechanical buttons require restrictions on input direction and preliminary operations such as pressing, thus having many restrictions on input operations. These are restrictions born from the judgment mechanism of physical switches and are not technically necessary to realize UIs in XR spaces. Furthermore, existing hardware is designed on the premise of high-precision tactile feedback, making it not optimal for current XR devices with limited haptic feedback. Therefore, the purpose of this research is to set out to achieve ease of mental model construction and operability optimized for XR environments.

In this study, we propose a spatial UI system that uses soap bubbles as a metaphor to facilitate the reuse of Interaction Knowledge and improve operability (Figures 1 and 2). The suitability of soap bubbles as a metaphor for spatial UIs lies in their three-dimensional nature, which is compatible with stereoscopic displays in XR environments, and their floating nature, which is compatible with the ability to negate gravity. As specific applications, we propose three types: "Bubble-Button" applied to button operations, "Bubble-Holder" applied to object movement operations, and "Bubble-Loading" applied to loading representations.

In this paper, among the three proposed applications, we conducted an experiment to compare the usefulness of Bubble-Button, which is likely to have a high impact on UI operation performance, with existing UIs used in XR. To compare overall performance in realistic situations, we performed pointing tasks with multiple button layouts. The experimental results revealed that Bubble-Button is faster in input speed, less tiring, and more attractive compared to Poke and Touch methods, but has a higher error rate.

2 Related Work

2.1 Button UIs

Poke-style button UIs [5] are often used in XR environment UIs. Poke-style is a pointing input UI in which the operation plane is pushed in with a finger, and selection occurs at the moment

the finger is released from the judgment plane after reaching the judgment plane. This feature reduces erroneous operations. On the other hand, preparatory actions increase, increasing the time the arm is held in the air [22], which can lead to fatigue [15].

These problems may be solved by applying Crossing input [7, 8, 13, 19] to VR [17, 26]. Tu et al.'s research shows that applying Crossing input to raycasting operations can complement or replace Pointing input [26].

2.2 Object Manipulation Techniques

In XR environments, there are many scenes of manipulating 3D objects for movement and placement [9, 21]. Techniques for manipulating objects located far away using raycasting from controllers [16] and Gaze and Pinch [23, 24, 27] are often seen. This operation has the problem that only one object can be moved per hand. Also, it is visually unclear whether an object released in mid-air will then float or fall. In the real world, objects released in mid-air fall, but in XR environments, how objects move depends on the application. Therefore, in both cases of falling and floating, it is easy for users' expectations to be betrayed.

2.3 Loading Displays

When using multiple applications simultaneously in an XR environment, it is necessary to asynchronously load content such as 3D models for performance reasons.

2D displays such as Spinners are often used as displays while 3D content is being loaded. For example, in visionOS loading displays, the `ProgressView()` function, which displays a Spinner, is prepared in advance as a standard OS function. Problems with 2D displays such as Spinners include that the shape of the content to be loaded cannot be predicted, and the display appears abrupt.

As a method to make it easier to understand the shape of the 3D object to be loaded, a Level of Detail (LoD) loading system is mainly used in games. However, with LoD, it is necessary to load a dedicated loading display model for each content, which has a high impact on performance. Therefore, it is difficult to use in complex resource management situations where multiple applications are launched simultaneously.

Skeleton Screen [28] is used in 2D UIs as a method with good performance and easy prediction of content size. This is claimed to make it easier to understand that loading is in progress and to have the effect of shortening the perceived loading time [28]. In a study by Mejtoft et al. [20] comparing perceived loading time between Spinner and Skeleton Screen, there was no significant difference, but results suggested that Skeleton Screen may make loading feel faster. Skeleton Screen is actually used in many services such as YouTube, Facebook, and Slack. However, there is no similar method in 3D or XR environments.

3 Bubble-UI

Bubble-UI is a spatial UI system that uses bubbles as a metaphor. Specifically, we propose three types: "Bubble-Button" applied to button operations, "Bubble-Holder" applied to object movement operations, and "Bubble-Loading" applied to loading representations.

The properties of soap bubbles that serve as guidelines when using them as a metaphor for spatial UIs are as follows:

Three-dimensional. Compatible with stereoscopic displays in XR environments.

Floating. The presence or absence of gravity can be visually represented. Therefore, it is compatible with the property of XR environments where physical laws can be freely set.

Transparent. Objects can be contained inside. Also, expressions can be matched with many styles, from photorealistic to illustrative.

Instantaneous generation and disappearance. Compatible with the properties of digital spaces where objects can be generated and disappeared instantaneously.

If utilizing these characteristics allows for the reuse of interaction knowledge or improves operability, adopting the metaphor of soap bubbles has its advantages. Properties of soap bubbles that should not be carried over can be based on criteria such as reduced operability or no user discomfort even if not expressed. For example, the property of moving in the wind is unnecessary because it makes position control and prediction difficult and reduces operability. Also, expressions that require operations unnecessary for realizing the operation, such as operations generated by blowing breath, are considered unnecessary.

3.1 Bubble-Button

Bubble-Button is a direct-touch operated, bubble-looking button with Crossing input. Input is judged the moment the user touches the soap bubble, and an effect of the soap bubble bursting is displayed (Figure 1). This method has no restrictions on input operations because there are no preliminary operations. Therefore, input is possible even with a swing-through motion that minimizes the time the arm is held in the air, and fatigue reduction is expected. Also, physical soap bubbles have little tactile sensation when touched. Therefore, discrepancies with expectations are less likely to occur even with devices with poor force feedback. Furthermore, the appearance of soap bubbles is thought to easily evoke the operation of touching directly with the hand.

3.2 Bubble-Holder

Bubble-Holder is a method of applying the soap bubble metaphor to scenes of manipulating object movement and placement (Figure 2). The metaphor is applied when moving objects using Gaze+Pinch or raycasting. Selected objects are wrapped in soap bubbles and begin to float in the air. The floating state is maintained even when released in mid-air. The floating state of an object can be canceled by momentarily tapping the fingertip while pointing at it. At the same time as the floating state is canceled, the soap bubble bursts, and the object inside begins to fall. With this specification, changes in state that are normally invisible, such as falling and floating, can be confirmed by the presence or absence of soap bubble display. Also, multiple objects can be grouped and moved. By bringing objects closer to each other while holding different objects in each hand, they can be grouped. This allows multiple objects to be moved simultaneously with one hand.

3.3 Bubble-Loading

Bubble-Loading displays an opaque soap bubble while a 3D object is being loaded (Figure 2). The soap bubble is generated in a size that

encloses the object. It bursts when the object loading is completed, and the object inside appears. Features include that the object size can be predicted and that the impact on performance is low. Since it is a soap bubble of a size that encloses the object, it is thought to be easy to predict how large the object will be and how it will be placed after loading. Also, since the same soap bubble model can be used for all loadings, the impact on performance may be low.

4 Experiment

In this experiment, we investigate the usefulness of Bubble-Button, which is most likely to affect UI operation performance among the three types of applications. To compare with existing methods, we investigated performance evaluation in multiple layouts and sizes, and workload and user experience evaluation by questionnaire. As existing methods for comparison, we used Poke-style and Touch-style buttons, which are commonly used in 3D UIs (Figure 3(a)). As tasks, we designed a task of selecting 12 targets arranged in a circle designed according to the ISO 9241-400:2007 [2] task, which is often used for pointing performance evaluation, and a task of inputting buttons in a grid arrangement close to actual usage scenes (Figure 3(b)). For workload evaluation, we used the Japanese version program of NASA-TLX [1, 14]. For user experience, we used the Japanese version of UEQ [25].

4.1 Participants and Apparatus

The participants were 12 (11 males, 1 female, average age = 21.8 years), aged 19 to 26 years. 11 participants were right-handed and 1 was left-handed. Participants were recruited using SNS within the affiliated university. All participants were unaware of the purpose of the experiment, had normal or corrected-to-normal vision, no sensory impairments, and used their right hand. None of the participants were inexperienced with VR, with 3 having used it 3-5 times, 1 about once a year, 4 about once every half year, 3 about once a month, and 1 using it daily.

The experimental apparatus consisted of a Meta Quest 3 head-mounted display (HMD) and a laptop computer (intel Core i7-12400, memory 32GB, NVIDIA GeForce RTX 3070Ti). The VR system used in this experiment was created on Unity (version 2024.3.14f1). Participants' avatars were not displayed in the VR space, only virtual hands as virtual bodies were displayed. Participants remained seated in chairs during the experiment. Answers to questionnaires were answered using a PC and iPad Pro. placed in positions that did not interfere with the experiment.

4.2 Experimental System

As the background of the VR space, we used a room model based on gray color included in Meta Interaction SDK. Poke-style and Touch-style buttons included in Meta Interaction SDK v59.00 were used. The pushing distance of the Poke-style was unified to 3cm. Penetration prevention processing to prevent the hand model from penetrating the button was applied to both Poke-style and Touch-style. These behaviors were set with reference to the sample scene of Meta Interaction SDK v59.00.

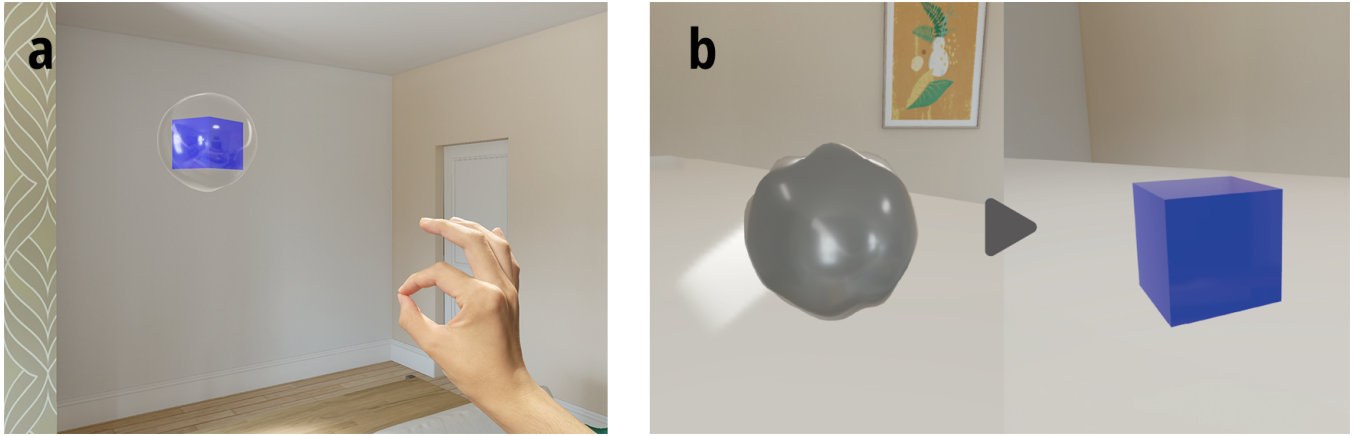


Figure 2: (a) Bubble-Holder: pinch to enclose an object in a bubble for levitation and dragging; brief pinch again to release. (b) Bubble-Loading: a bubble appears during content loading and bursts when finished, revealing the object.

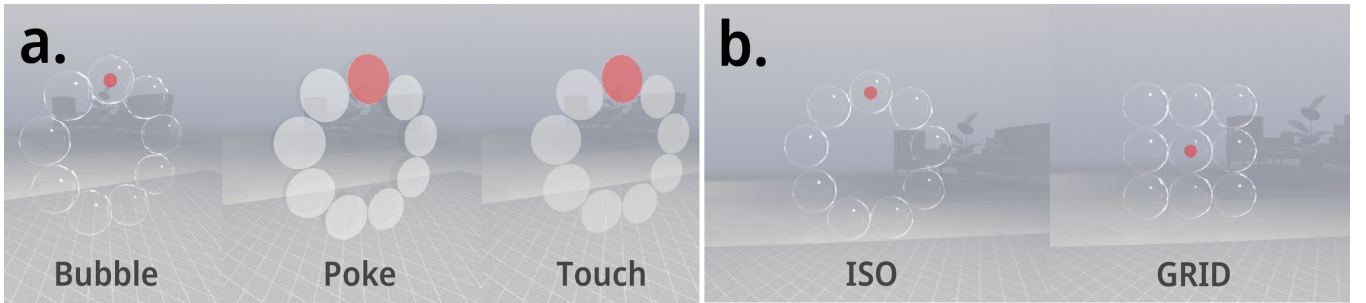


Figure 3: (a) Types of buttons used (*type*). Task is to select the red highlighted target. (b) Layouts of buttons used (*layout*). In ISO, targets along the diagonal are selected sequentially, whereas in GRID, targets are chosen randomly without overlap.

4.3 Task

In this experiment, we performed a task of selecting 12 targets arranged in a circle (ISO) designed according to the guidelines of ISO 9241400:2007 [2] and a task of arranging targets on a 3x3 Grid (GRID). Targets arranged as shown in Figure 3 were placed 45 cm ahead of the user on a vertical plane, and participants were able to select targets while seated. Selection was performed using the index finger, and in ISO, the selection order was determined in advance so that the distance between targets was constant in all trials. In GRID, the selection order was set randomly, and in one trial, it was set so that all targets were selected once each without duplication. Selection is completed by touching and releasing the target with the fingertip for Poke and Touch, and automatically by touching for Bubble. This is in accordance with the general implementation of Meta Interaction SDK. Targets to be selected by participants were highlighted in red to avoid confusion, and other targets were set to white. When the target to be aimed at was selected, the color of the next target to be aimed at changed.

Each trial of the task was performed with different target sizes and distances between targets. Target sizes were performed in two types: 1) small (diameter = 5cm), 3) large (diameter = 10cm). Distances between targets were set in 3 types each for ISO and

GRID. For ISO, 1) short = 30cm, 2) medium = 45cm, 3) large = 60cm. For GRID, 1) short = 10cm, 2) medium = 15cm, 3) large = 20cm.

4.4 Procedure

After explaining the details of the experiment in advance, consent forms for participation in the experiment were obtained from the subjects. This experiment used a quiet laboratory with minimal interference from external acoustic stimulation and temperature changes to verify the effect of visual feedback. Next, we provided guidance on wearing the VRHMD, basic operations, and initial position calibration methods. On top of that, we explained the content of the tasks in the virtual environment and provided a practice session of about 60-120 seconds to get used to the operations before starting the tasks. For the tasks, participants were told to "select the targets as quickly and accurately as possible."

Experimental blocks were nested in the order of layout, type, size, and distance. Blocks in each layer were randomized for each participant to eliminate order effects. The total number of trials for the task was 108 trials of 2 layouts (*layout*) $\times$ 3 button types (*type*) $\times$ 2 button sizes (*size*) $\times$ 3 distances between buttons (*distance*) $\times$ 3. NASA-TLX and UEQ questionnaires were conducted 6 times each time the *layout* block was completed.

5 Results

The results of task performance and questionnaires measured in this experiment are shown below. RM-ANOVA was performed for statistical analysis of task performance to verify main effects and significant differences between conditions. As a procedure, normality was examined using the Shapiro-Wilk test, and then homoscedasticity was confirmed by Mauchly's sphericity test. If sphericity was not satisfied, p-values were corrected using Greenhouse-Geisser, and Bonferroni multiple comparisons were performed in post-hoc tests. RM-ANOVA was used even though it was non-parametric data. The reason is that RM-ANOVA has higher statistical power compared to non-parametric tests and is robust against violations of normality [10]. For statistical analysis of questionnaires, the Friedman test was performed. For pairwise comparisons, the Wilcoxon signed-rank test was used, and Bonferroni correction was applied.

5.1 Number of Errors

The number of errors for each *type* is shown in Figure 4(a). In the GRID layout, a significant main effect was shown in *type* ($F_{1.011,11.122} = 6.184, p = .030, \eta^2 = .360$). In the ISO layout, significant main effects were shown in *type* ($F_{1.015,11.166} = 14.314, p = .003, \eta^2 = .565$) and *distance* ($F_{2,22} = 8.720, p = .002, \eta^2 = .442$).

5.2 Input Time

The results of input time are shown in Figure 4(b). In the GRID layout, significant main effects were shown in *type* ($F_{2,22} = 76.39, p < .001, \eta^2 = .874$), *size* ($F_{1,11} = 48.580, p < .001, \eta^2 = .815$), and *distance* ($F_{2,22} = 107.034, p < .001, \eta^2 = .907$). In the ISO layout, significant main effects were shown in *type* ($F_{2,22} = 27.92, p < .001, \eta^2 = .717$), *size* ($F_{1,11} = 89.926, p < .001, \eta^2 = .891$), and *distance* ($F_{1,285,14.134} = 127.358, p < .001, \eta^2 = .920$).

5.3 NASA-TLX

The questionnaire results of NASA-TLX are shown in Figure 4(c). The results of Friedman test for *type* in each *layout* condition are as follows. In the GRID layout, significant differences were shown in Physical Demand ($\chi^2(2) = 13.3778, p = .0012$), Temporal Demand ($\chi^2(2) = 7.5135, p = .0234$), Frustration ($\chi^2(2) = 15.2444, p = .0005$), and Overall ($\chi^2(2) = 13.1667, p = .0014$). In the ISO layout, significant differences were shown in Physical Demand ($\chi^2(2) = 18.7273, p < .001$), Effort ($\chi^2(2) = 11.9111, p = .0026$), Frustration ($\chi^2(2) = 11.6522, p = .0029$), and Overall ($\chi^2(2) = 16.1667, p < .001$).

5.4 UEQ

The questionnaire results of UEQ are shown in Figure 4(d). The results of Friedman test for *type* in each *layout* condition are as follows. In the GRID layout, significant differences were shown in Attractiveness ($\chi^2(2) = 11.0213, p = .0040$), Efficiency ($\chi^2(2) = 6.3182, p = .0425$), Stimulation ($\chi^2(2) = 11.6170, p = .0030$), and Novelty ($\chi^2(2) = 18.7826, p = .0001$). In the ISO layout, significant differences were shown in Attractiveness ($\chi^2(2) = 14.0851, p = .0009$), Efficiency ($\chi^2(2) = 7.3478, p = .0254$), Stimulation ($\chi^2(2) = 15.1667, p = .0005$), and Novelty ($\chi^2(2) = 20.4681, p < .0001$).

6 Discussion

The main reasons why Bubble-Button shortened input time compared to other methods are that preliminary operations are unnecessary and the degree of freedom in input operations is high. Poke-style requires a preliminary operation of pushing in the button, and Touch-style also requires an operation of touching and releasing [22]. Also, Bubble-Button has a larger surface area even with the same radius size, so it is thought that the hit judgment was large and easy to touch.

From the NASA-TLX results, it was shown that Bubble-Button has significantly lower Physical Demand, Effort, Frustration, and Overall compared to other methods. As a specific quantity, Overall was about 30% lower. This is thought to be because the physical load on the user was reduced due to the reduction of preliminary operations and the high degree of freedom in input operations.

The UEQ results showed high scores for Attractiveness, Stimulation, and Novelty, suggesting that the quality of user experience is improved. In user interviews, there were multiple responses such as "I want to touch it." The bubble metaphor may have provided freshness and fun, and increased motivation for operation.

On the other hand, it was revealed that Bubble-Button has a problem of high error rate. The reason for the high error rate especially in the ISO layout is thought to be that the possibility of accidentally touching adjacent buttons increased when the distance between buttons was short. The reason why there was no significant difference in the number of errors in GRID is thought to be that the bias in the number of errors was large among participants. In GRID, a difference in the number of errors was seen between some participants who were proficient in VR operations and those who were not. As a countermeasure, it is conceivable to use it in combination with a confirmation modal for important selections. Also, it is considered appropriate to use it in scenes where re-input is allowed.

From these results, Bubble-Button is expected to be utilized in XR applications that require long hours of work, rehabilitation, etc. For example, in OS setting screens, design support tools that require frequent menu operations and object operations, and menus in XR games, etc., in situations where redo is possible even if wrong input is made, it has the potential to reduce user burden and improve productivity.

Also, as an attempt to further utilize the soap bubble metaphor, there is a possibility of helping depth perception by reflection. Specifically, by incorporating reflection expressions that reflect the user's hands and surrounding environment in real time on the bubble surface, it may become easier to visually grasp the relative position to the bubble, contributing to reduction of erroneous operations and improvement of operation accuracy.

In this study, we focused on the evaluation of Bubble-Button, and did not evaluate Bubble-Holder and Bubble-Loading. It is necessary to conduct usability evaluations for these methods as well and verify their comprehensive usefulness.

Furthermore, by conducting qualitative evaluations, it may be possible to clarify the influence of metaphors on users' mental model formation. In particular, clarifying how users accept the soap bubble metaphor, what kind of changes occur in mental model



Figure 4: Result plots: (a) Box plot of Number of Errors, (b) Box plot of Input Time (seconds), (c) Box plot of NASA-TLX score (lower score is better), (d) Box plot of UEQ score (higher score is better). Significant differences in pairwise comparisons are indicated by *, **, and *, representing Bonferroni-adjusted significance levels of 0.05, 0.01, and 0.001, respectively.**

construction, and how that affects operations will lead to theoretical contributions to the design of metaphors used in UIs.

7 Conclusion

In this study, we proposed a spatial UI system "Bubble-UI" using soap bubbles as a metaphor and examined its usefulness. Specifically, we proposed three types: "Bubble-Button" applied to button operations, "Bubble-Holder" applied to object movement operations, and "Bubble-Loading" applied to loading representations.

As a result of experiments, Bubble-Button was faster in input speed, less tiring, and highly evaluated in terms of user experience compared to existing Poke-style and Touch-style. On the other hand, the problem of high error rate was also revealed. This is thought to be because erroneous input is likely to occur because the judgment range is wide and there are no restrictions on the input direction. To solve this problem, it is necessary to make improvements such as adjusting the spacing between buttons.

It was suggested that by using the soap bubble metaphor, it is possible to construct operability optimized for XR environments and intuitive mental models. Future issues include conducting performance evaluations and usability evaluations of Bubble-Holder and Bubble-Loading, which were not evaluated this time. It is also necessary to conduct more qualitative evaluations on the influence of metaphors on mental models.

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